

# Management Tools and Architectures

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## Variety of tools

Reasons for this:

- Diversity of resources to be administered
- Various characteristics of a resource
- Different comprehensive and complex IT infrastructures: PC network, corporate network, carrier network
- Different requirements for the tools from different user views
- New technologies often require special tools
- Tools are often manufacturer specific

## Variety of tools - 2

- **Network configuration tools**
  - Ipconfig, ifconfig, iwconfig, ethtool, mii-tool, route, arp, netStumbler, kismet, ranchid, tools to manage VLANs, etc.
- **Network benchmark, measurement and estimation tools**
  - Ping, tcpping, owamp, lperf, netperf, pathrate, pathload, httpperf, Apache Benchmark Tool, etc.
- **Network statistics and status monitoring tools**
  - Cacti, Nagios, MRTG, NetFlow, argus, PRTG Network Monitor, Pandora FSM, RRDTool, etc.
- **Security tools**
  - Nmap (port scanner), Nessus (vulnerability scanner), Snort (intrusion detection system), firewalls (iptables), etc.
- **Backup tools**
  - Acronis True Image, Paragon Backup and Recovery, Windows/Mac/Unix backup solutions, virtual machine image backups, etc.

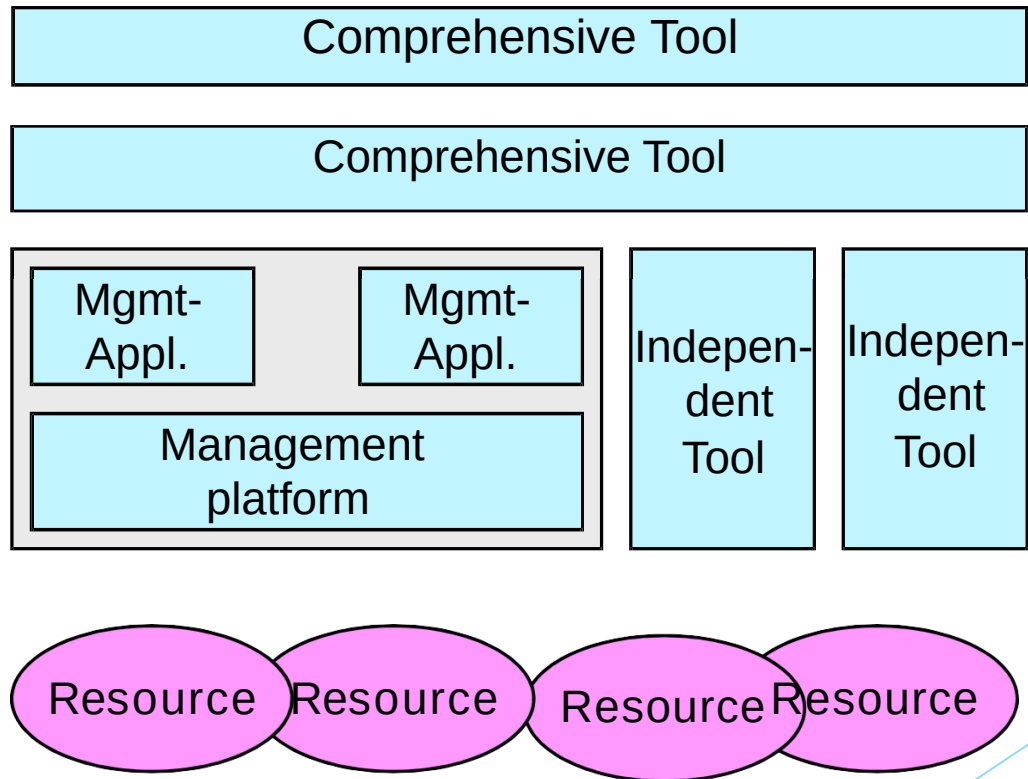
## Variety of tools - 3

- **CMDB tools**
  - i-doit, Micro Focus Universal CMDB, ServiceNow CMDB, IBM Tivoli CMDB, BMC Helix CMDB, openCMDB, etc.
- **Development tools**
  - MIB tools, agent development, application development, interface tools, etc.
- **Protocol analyzers**
  - Wireshark, tcpdump, gulp, etc.
- **Network management platforms**
  - HP Openview Platform, Patrol Management System, Tivoli TME Management Platform, Cabletron Spectrum Management Platform, etc.

## Possible classification dimensions for tools

- Structure and components of the tool
  - Pure software solution
  - Pure HW solution
  - Combination HW / SW solution
- Type of realization:
  - Distributed realization
  - Local realization
- Covered task area
- Autonomy of the tools
  - Stand-alone tools
  - Platforms
  - Comprehensive tools

## Tool architecture according to the dimension of autonomy



## Dimension of independence (1)

### Independent tools

- can be operated without relying on another tool
- are aimed at a specific management goal

### Platforms

- can be operated independently, but do not provide any or only insufficient functionality (e.g. MIB browsing)
- Platforms include accessibility, graphical user interface, persistent storage of management information databases, management application APIs
- Tight coupling with the management applications running on the platform

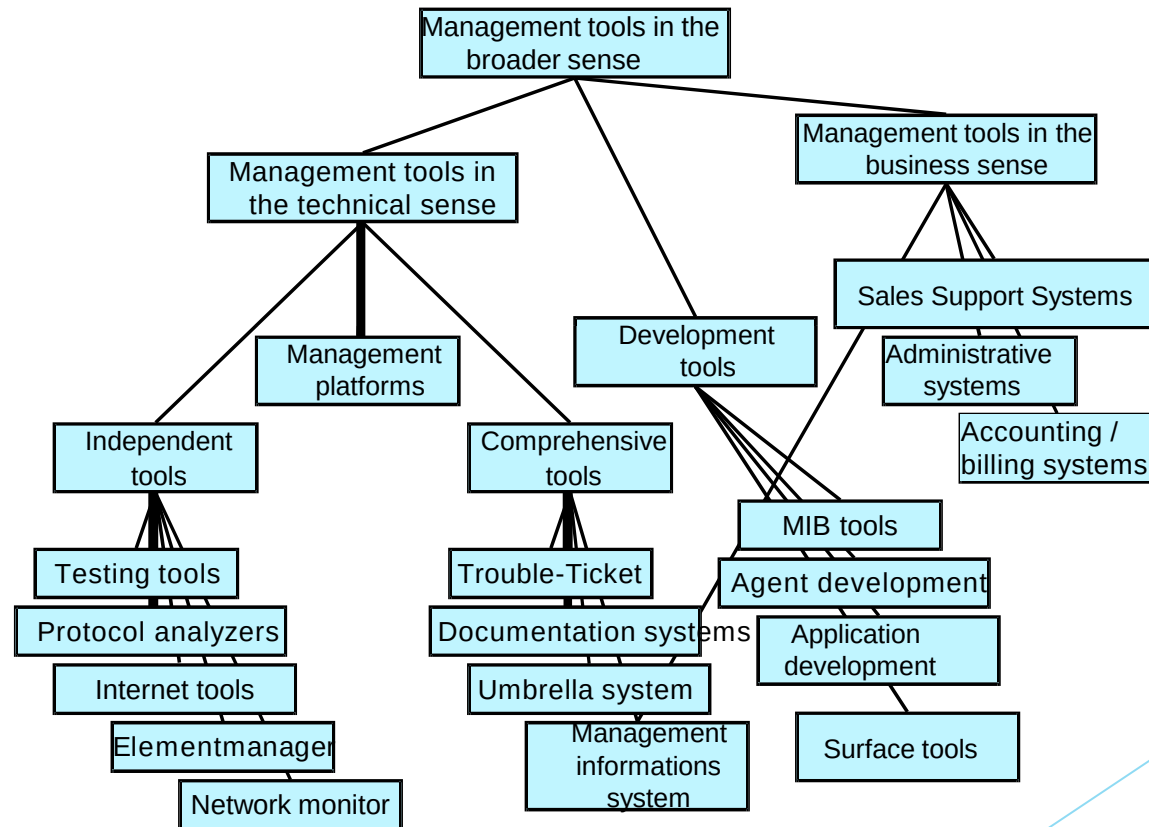
## Dimension of independence (2)

### Comprehensive tools

- can be operated formally independently, but then make no sense
- rely on other tools or integrate other tools (Umbrella system)
- Merging and processing and distribution of management information
- Loose coupling with the other management tools



## Classification scheme for management tools



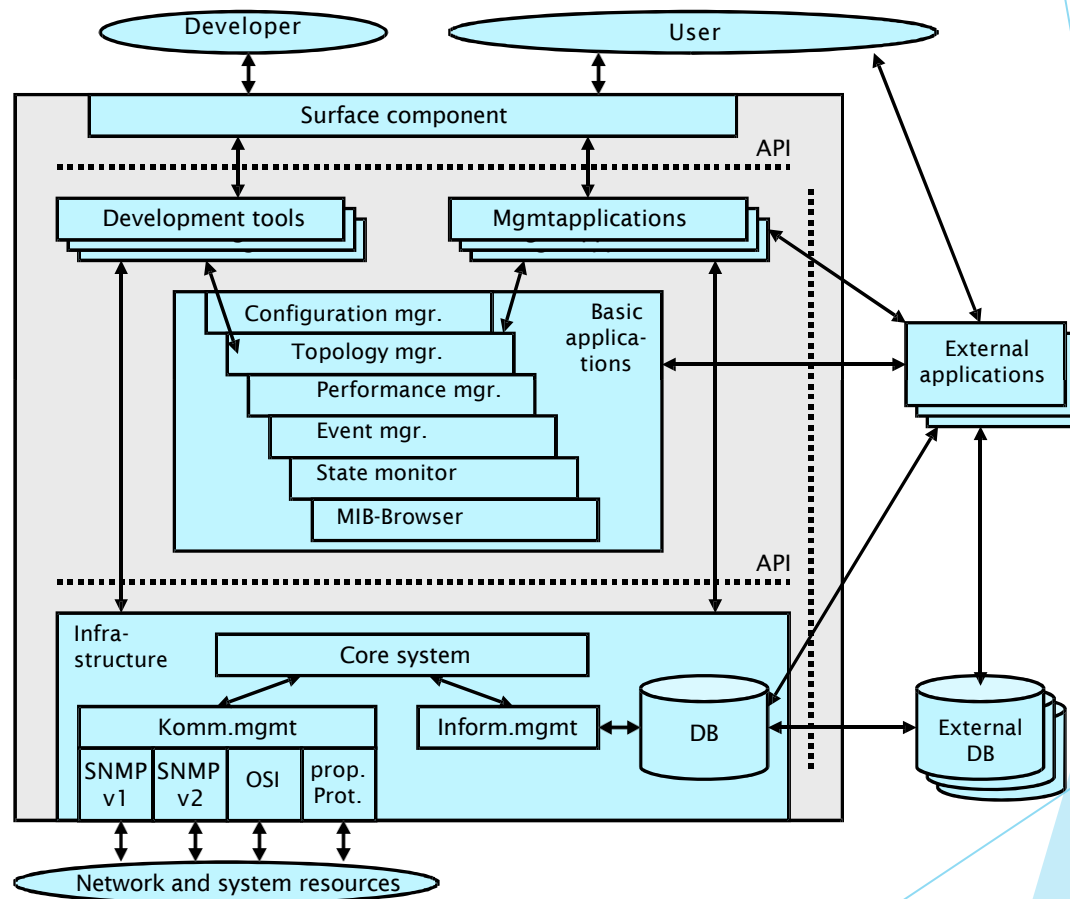
## Systematic selection of tools

- Requirements Analysis: Overall Concept IT Structure. What should be achieved? User requirements, business goals, specifications
- Is analysis: Network structure, administration structure, management tools
- Target concept: network structure, administration structure, management tools
- Implementation alternatives : Network structure (product strategy) → Administration ( outsourcing yes / no / partial) → Architecture of management tools (platform / individual tools )
- Product choice: Evaluation of the tools (technical, architectural (interfaces, integration into the overall concept, extensibility), organizational (operating concept, integration into the process structure), business management (investment protection, purchase price, reputation manufacturer, costs after sales ...))

## Stages of tool integration

- Isolated tool integration
- Coordinated tool integration
  - Surface integrated tools
  - Script-integrated tools (loosely coupled)
- Platform integrated approach  
(Integration via information, organization, communication and function model)

## Architecture of platforms



## Features and characteristics of platforms

Platform as an open carrier system for management applications:

- Open programming interfaces
- Open platform databases
- Comprehensive and open protocol support
- Uniform, open runtime environment
- Uniform, vendor-independent and open information model,
- Uniform, open user interface,
- Comprehensive and open basic functionalities (basic applications),
- Development environment

## Selection criteria for platforms

- Flexibility  
Open interfaces, surface adaptability, modularity
- Scalability  
Scalability may require that the platform itself be distributed ( -> CORBA).
- Integration with existing management and information environments
- Operational safety: data security, (wantonly) incorrect operation)
- Simplicity: Different user groups, training effort
- Cost / benefit ratio  
Cost: hardware / software, customization, training, operation.  
Benefits: quantitative / qualitative benefits
- Future security  
Stability of the manufacturer, third party for the platform
- Supported platform services and basic applications

## Basic applications / functions of platforms (1)

- Event management
  - External events / internal events
  - Event message (event type, trigger, priority, timestamp, network address)
  - Further processing by filter mechanism coupled with logs
  - Triggering actions
  - Intelligent Preprocessing of Events: Event correlation method with the goal of event reduction. → Avoiding event avalanches that lead to inefficient management (minimum requirement: prioritization)
  - Visualization of events (status window, integrated into the map (color of the icon))
- Configuration management
  - Dialing into systems (telnet via platform)
  - Manipulation of configuration via management protocols (SNMP)
  - Configuration change via manufacturer-specific modules (integrated in the surface of the platform)

## Basic applications / functions of platforms (2)

- Network Topology Management
  - Autodiscovery function
  - Autotopology function: Automatic construction of a map
  - Implementation via "learning bridges", ICMP (tracert), ARP, SNMP MIB access, access to manufacturer MIBs
- Performance monitoring
  - Measurements of response times, throughput, resource utilization
  - Management of measuring procedures (start, stop, ..)
  - Evaluation of the measurement results (statistical means, ...)



## Management applications (stand-alone or integrated in platforms)

- **Trouble Ticket and Help Desk**  
Technology: Filter Modules & Correlation Modules, Knowledge Base, Diagnostic Module  
Operating Environment: Automatic Call Distribution, Computer Telephone Integration, process optimization
- **Inventory Management**  
includes technical and business data of components, addresses, software
- **Facility Management**  
Cabling, building infrastructure (adapted CAD system)
- **Accounting systems**  
Collection and processing of billing-relevant information
- **Software distribution**  
Level of automation of distribution: file transfer, distribution, configuration  
Additional functions: version monitoring, license monitoring
- **Security Management**
- **Identity Management**
- **Network management**
- e.g. technology-related applications (e.g., VLAN), component management

# Network Monitoring Systems

# Multi Router Traffic Grapher (MRTG)

- The Multi Router Traffic Grapher (MRTG) is a tool to monitor and measure the traffic load on network-links.
- MRTG allows the user to see traffic load on a network over time in graphical form“.
- MRTG generates HTML pages containing PNG images which provide an almost live visual representation of this traffic.
- MRTG capitalizes on the Simple Network Management Protocol, or SNMP, a feature that is built into most network equipment.
- SNMP-enabled devices have counters that can be remotely read by MRTG.

# Multi Router Traffic Grapher (MRTG) (2)

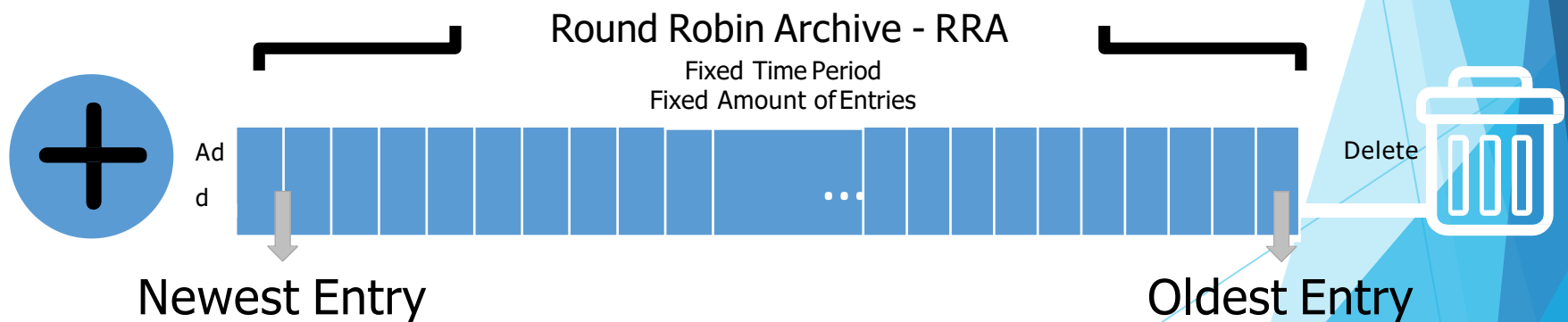
- External readers for MRTG graphs can create other interpretation of data.
- MRTG software can be used not only to measure network traffic on interfaces, but also build graphs of anything that has an equivalent SNMP MIB - like CPU load, Disk availability, Temperature, etc...
- Data sources can be anything that provides a counter or gauge value – not necessarily SNMP.
  - For example, graphing round trip times
- MRTG can be extended to work with RRDTool

# MRTG - Issues

- MRTG generates each graph (what if you have hundreds of graphs!) every 5 minutes, creating a lot of overhead.
- It also has very few customizable graphing options.
- Disk space is always an issue.
- MRTG is a rather old tool with a strong Unix background. However, it has also been ported to the Windows platform a while ago.
- Configuration of MRTG is mostly done via text files that must be edited. No GUI control panels and autoconfiguration wizards
- In MRTG, there is no discovery process that will scan your network for SNMP manageable devices and let you pick those you want to monitor.

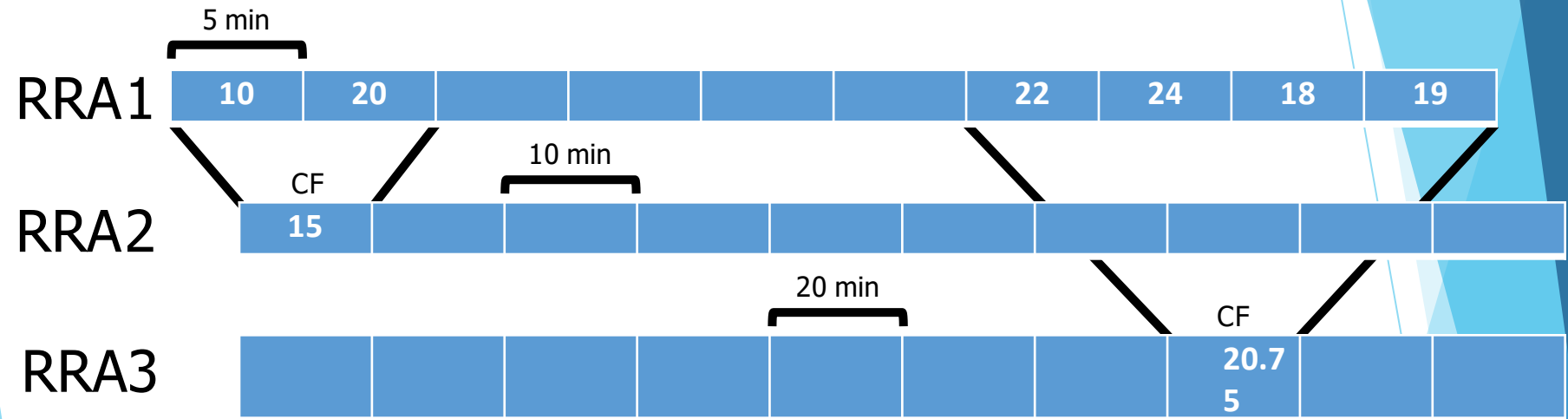
# What is RRDTool?

- ▶ Round Robin Database for time series data storage
- ▶ From the author of MRTG
- ▶ Data is polled from target machine(s) and stored into different Round Robin Archives (RRAs) that exist within a single Round Robin Database (RRD) file.
- ▶ RRAs have a predefined amount of space for data that overwrites itself as it becomes full.
- ▶ Therefore, the RRD never grows larger in filesize.



Once the archive is full, when new data is added and the oldest entry is destroyed

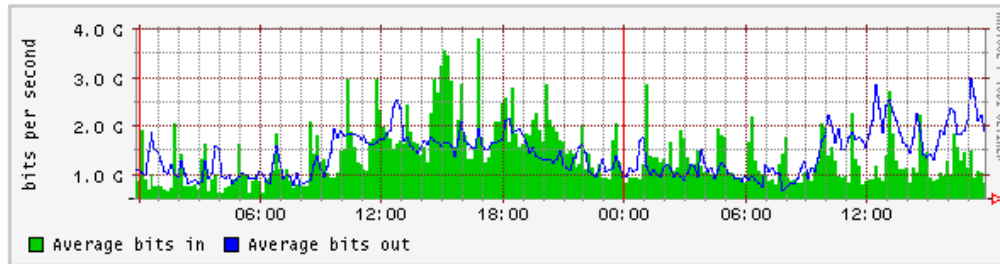
# Round Robin Archives (RRAs)



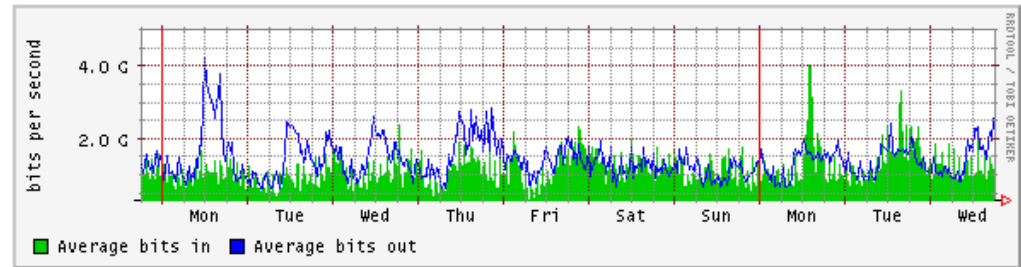
- Multiple RRAs can be used to provide consolidated data over different time frames.
- RRDTool can then read from the specified RRD files to create a graph

# RRD Tool for MRTG graphs

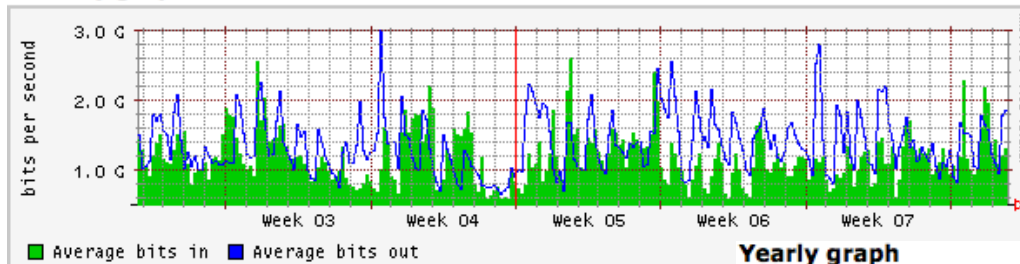
**Daily graph**



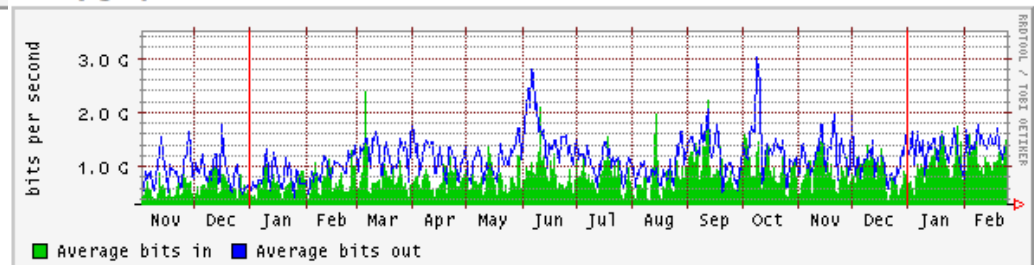
**Weekly graph**



**Monthly graph**



**Yearly graph**





# Cacti (1)

**Cacti:** Uses RRDtool, PHP and stores data in MySQL. It supports the use of SNMP and graphics with MRTG.

*“Cacti is a complete frontend to RRDTool, it stores all of the necessary information to create graphs and populate them with data in a MySQL database. The frontend is completely PHP driven. Along with being able to maintain Graphs, Data Sources, and Round Robin Archives in a database, cacti handles the data gathering. There is also SNMP support for those used to creating traffic graphs with MRTG.”*



# Cacti (2)

- Cacti is a close parent of MRTG.
- Cacti is a complete network graphing solution designed to harness the power of RRDTool's data storage and graphing functionality.
- Almost all of Cacti's functionality can be configured via the Web.
- Cacti also features user access control built right into the product. It's also got an easy to use web-based interface.
- ▶ One of its strongest features is its templates. There is a built-in template, for example, for Cisco switches that already includes most of the elements you might want to monitor on such a device.
- ▶ And you can build device-specific templates if they don't already exist.
- ▶ There are also many device-specific templates that can be downloaded either from each device manufacturer's websites or from community Cacti forums.

# Nagios and Availability Management

- Availability Management is a process of the Service Delivery (ITIL specifications). Its goal is to ensure that the required level of Availability is provided.
- The measurement and monitoring of IT Availability is a key activity to ensure availability levels are being met consistently.
- In a network management environment, monitoring services needed are for example:
  - Routers status
  - Routers environment (CPU, Memory, Temperature...)
  - Links status (interconnection, access)
  - Links health (traffic evolution, errors level, instability...)
  - Routing protocol status (BGP, MPLS...)
  - ...
- The functionalities needed are:
  - Extended monitoring capabilities, including custom programs for non-standard elements
  - Powerful analysis and notification when an incident is detected
  - Easy to use configuration interface
  - Friendly GUI for service status display

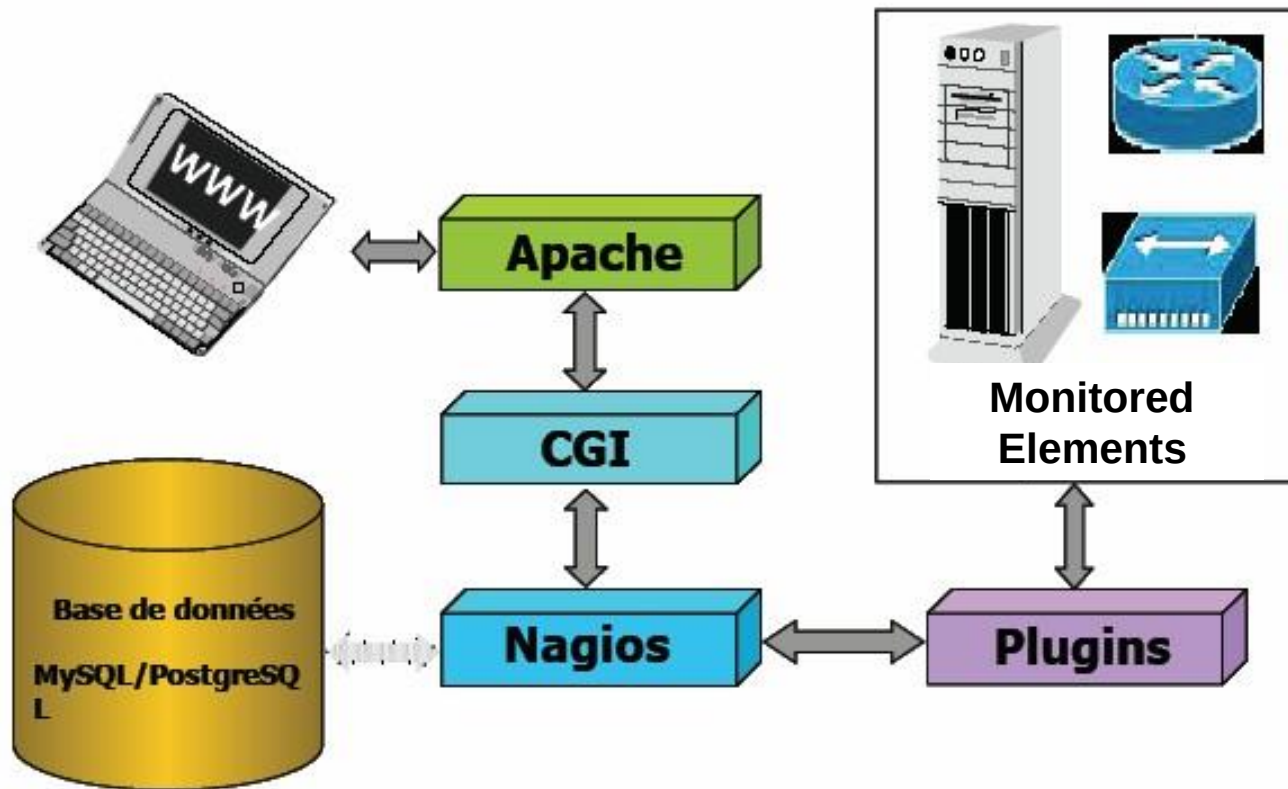
# Nagios functionalities (1)

- Nagios® is an open source tool specially developed to monitor the availability of hosts and services, and designed to inform you of network incidents before your clients, end-users or managers do.
- Nagios works with a set of “plugins” to provide local and remote service status. The monitoring daemon runs intermittent checks on hosts and services you specify using external "plugins" which return status information to Nagios.
- When incidents are detected, the daemon send notifications out to administrative contacts in a variety of different ways (email, instant message, SMS, etc.).
- Current status information, historical logs, and reports can all be accessed via a Web browser.
- Custom “plugins” are relatively easy to develop
- Different methods are provided for remote resource discovery
- Nagios is freely available from <http://www.nagios.org>

# Nagios functionalities (2)

- Nagios tool provide monitoring facilities (polling elements at regular intervals and generating events depending on polling return status.
- Nagios can be considered as a scheduler, which execute plugins, analyse results and obtain a status about each element concerned.
- Nagios can be configured to monitor anything that you can write a script for (Perl scripts, Shell, C programs, ...).
- Nagios comes with loads of plugins for just about every normal known service such as SNMP, DNS, SSH, HTTP, NFS, FTP, etc, etc.
- The advantages of Nagios against other manufacturer products are:
  - It is a free product, running on Linux OS
  - It is an easy to use product, including all functions needed to perform availability monitoring
  - Extensions (plugins) are simple to develop, using well known languages
  - It is scalable, and it is possible to deploy a hierarchy of Nagios servers to provide consolidated views of network and / or system availability (Hypervisor)

# Nagios architecture





il screen

General

- Home
- Documentation

Monitoring

- Tactical Overview
- Service Detail
- Host Detail
- Status Overview
- Status Summary
- Status Grid
- Status Map
- 3-D Status Map
- Service Problems
- Host Problems
- Network Outages
- Comments
- Downtime
- Process Info
- Performance Info
- Scheduling Queue

Reporting

- Trends
- Availability
- Alert Histogram
- Alert History
- Alert Summary
- Notifications

Current Network Status

Last Updated: Sun Feb 1 12:17:48 NPT 2004  
Updated every 90 seconds  
Nagios® - [www.nagios.org](http://www.nagios.org)  
Logged in as *dhruba*

[View Service Status Detail For All Host Groups](#)  
[View Status Overview For All Host Groups](#)  
[View Status Summary For All Host Groups](#)  
[View Status Grid For All Host Groups](#)

Display Filters:

Host Status Types: All problems  
Host Properties: Any  
Service Status Types: All  
Service Properties: Any

Host Status Totals

| Up  | Down | Unreachable | Pending |
|-----|------|-------------|---------|
| 155 | 15   | 0           | 0       |

All Problems

15

All Types

170

Service Status Totals

| Ok  | Warning | Unknown | Critical | Pending |
|-----|---------|---------|----------|---------|
| 226 | 5       | 0       | 16       | 0       |

All Problems

21

All Types

247

Host Status Details For All Host Groups

| Host                           | Status | Last Check          | Duration       | Status Information                 |
|--------------------------------|--------|---------------------|----------------|------------------------------------|
| <a href="#">CHILDREN-FIRST</a> | DOWN   | 02-01-2004 12:13:59 | 1d 19h 10m 33s | PING CRITICAL - Packet loss = 100% |
| <a href="#">DANIDA</a>         | DOWN   | 02-01-2004 12:15:55 | 1d 0h 43m 12s  | PING CRITICAL - Packet loss = 100% |
| <a href="#">DASS</a>           | DOWN   | 02-01-2004 12:08:59 | 4d 0h 40m 42s  | PING CRITICAL - Packet loss = 100% |
| <a href="#">FNCCI</a>          | DOWN   | 02-01-2004 12:12:38 | 4d 0h 40m 2s   | PING CRITICAL - Packet loss = 100% |
| <a href="#">ITLINK</a>         | DOWN   | 02-01-2004 12:15:55 | 0d 1h 37m 12s  | PING CRITICAL - Packet loss = 100% |
| <a href="#">Laz-cnet</a>       | DOWN   | 02-01-2004 12:12:38 | 4d 0h 38m 53s  | PING CRITICAL - Packet loss = 100% |



# Web Interface: Service Types

Current Service Status - Mozilla

File Edit View Go Bookmarks Tools Window Help

https://thuldai.mos.com.np/nagios/cgi-bin/status.cgi?host=all Search

| Host    | Service     | Status | Time                | Duration       | Attempts | Details                                                                                |
|---------|-------------|--------|---------------------|----------------|----------|----------------------------------------------------------------------------------------|
| Kailash | Cpu-usage   | OK     | 02-01-2004 10:21:38 | 3d 22h 48m 34s | 1/3      | SNMP OK: usr-cpu:1, sys-cpu:1,                                                         |
|         | FTP         | OK     | 02-01-2004 10:23:48 | 3d 22h 46m 38s | 1/3      | FTP OK - 0.007 second response time port 21 [220 kailash.mos.com.np FTP server ready.] |
|         | Free-Memory | OK     | 02-01-2004 10:22:15 | 3d 22h 48m 34s | 1/3      | SNMP OK: Ram-Free:3100,                                                                |
|         | HTTP        | OK     | 02-01-2004 10:22:59 | 3d 22h 46m 38s | 1/3      | HTTP ok: HTTP/1.1 200 OK - 0.021 second response time                                  |
|         | Load        | OK     | 02-01-2004 10:25:17 | 3d 22h 48m 34s | 1/3      | SNMP OK: 1MIN-Load:0.08, 5MIN-Load:0.05, 15MIN-Load:0.00,                              |
|         | Ping        | OK     | 02-01-2004 10:25:07 | 0d 5h 7m 33s   | 1/3      | PING OK - Packet loss = 0%, RTA = 0 ms                                                 |
| Karnali | disk_usage  | OK     | 02-01-2004 10:22:51 | 3d 22h 48m 34s | 1/3      | Disk utilization: All disks OK                                                         |
|         | Ping        | OK     | 02-01-2004 10:25:58 | 0d 17h 48m 53s | 1/3      | PING OK - Packet loss = 0%, RTA = 1 ms                                                 |
| Kopila  | Cpu-usage   | OK     | 02-01-2004 10:24:07 | 3d 22h 48m 34s | 1/3      | SNMP OK: usr-cpu:0, sys-cpu:1,                                                         |
|         | Free-Memory | OK     | 02-01-2004 10:22:51 | 3d 22h 46m 38s | 1/3      | SNMP OK: Ram-Free:3808,                                                                |
|         | Load        | OK     | 02-01-2004 10:22:18 | 3d 22h 48m 34s | 1/3      | SNMP OK: 1MIN-Load:0.18, 5MIN-Load:0.19, 15MIN-Load:0.18,                              |
|         | POP         | OK     | 02-01-2004 10:23:07 | 3d 22h 46m 38s | 1/3      | POP OK - 0.028 second response time port 110 [+OK <8832.1075610415@kopila.mos.com      |
|         | Ping        | OK     | 02-01-2004 10:25:58 | 3d 15h 7m 15s  | 1/3      | PING OK - Packet loss = 0%, RTA = 1 ms                                                 |
| Koshi   | Ping        | OK     | 02-01-2004 10:22:37 | 1d 13h 37m 43s | 1/3      | PING OK - Packet loss = 0%, RTA = 9 ms                                                 |

Done

Mozilla-bi [root@dhr] ? Sun Feb 01, 9:56 PM



# Web Interface: Service Detail of Nagios

Current Service Status - Mozilla

File Edit View Go Bookmarks Tools Window Help

https://thuldai.mos.com.np/nagios/cgi-bin/status.cgi?host=all Search

**Current Network Status**  
Last Updated: Sun Feb 1 09:57:47 NPT 2004  
Updated every 90 seconds  
Nagios® - [www.nagios.org](http://www.nagios.org)  
Logged in as *dhruba*

[View History For all hosts](#)  
[View Notifications For All Hosts](#)  
[View Host Status Detail For All Hosts](#)

**Host Status Totals**

|     |      |             |         |
|-----|------|-------------|---------|
| Up  | Down | Unreachable | Pending |
| 155 | 15   | 0           | 0       |

|              |           |
|--------------|-----------|
| All Problems | All Types |
| 15           | 170       |

**Service Status Totals**

|     |         |         |          |         |
|-----|---------|---------|----------|---------|
| Ok  | Warning | Unknown | Critical | Pending |
| 228 | 3       | 0       | 16       | 0       |

|              |           |
|--------------|-----------|
| All Problems | All Types |
| 19           | 247       |

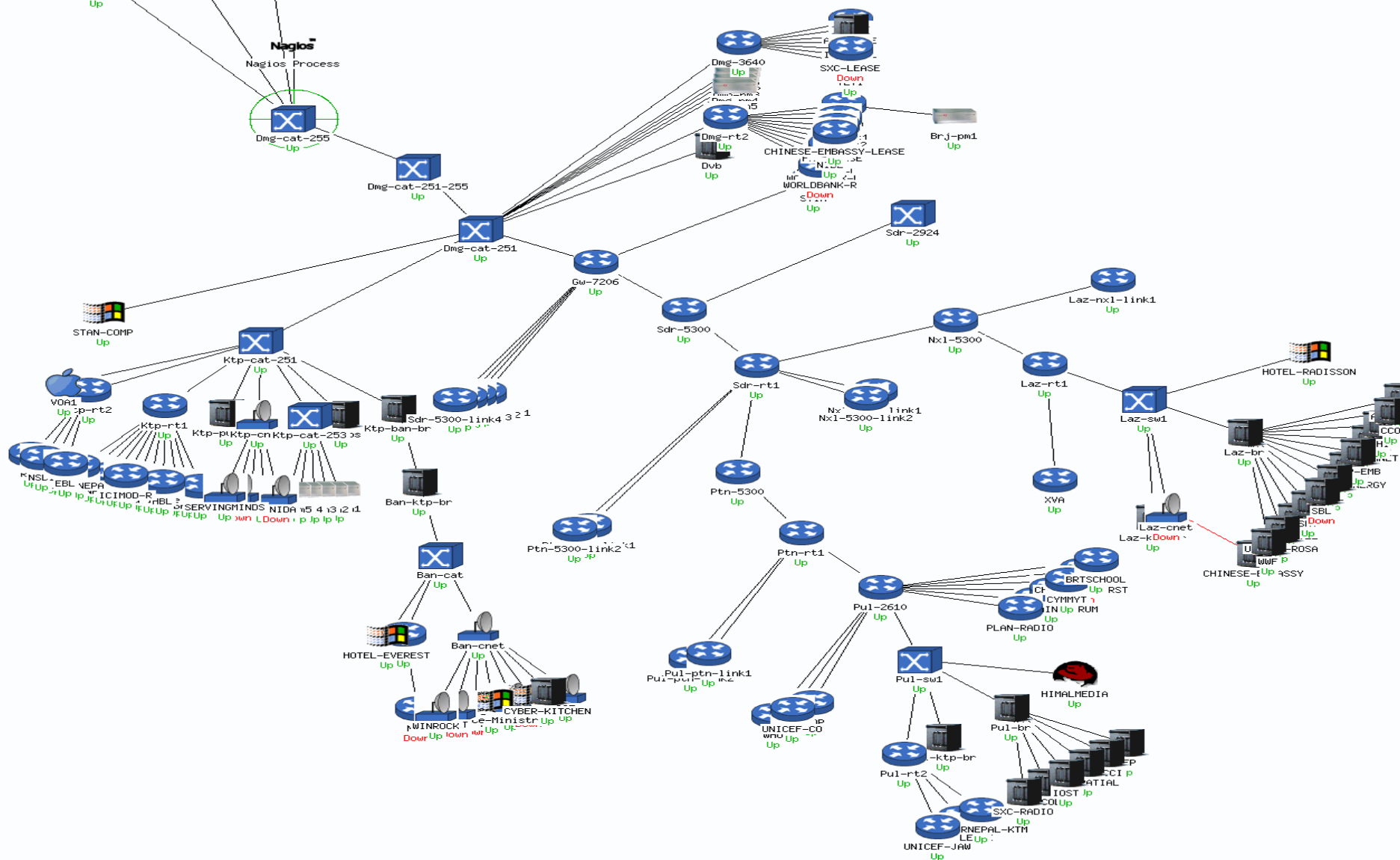
?

**Service Status Details For All Hosts**

| Host ↑↓                   | Service ↑↓ | Status ↑↓ | Last Check ↑↓       | Duration ↑↓    | Attempt ↑↓ | Status Information                    |
|---------------------------|------------|-----------|---------------------|----------------|------------|---------------------------------------|
| <a href="#">ACTIONAID</a> | Ping       | OK        | 02-01-2004 09:53:07 | 0d 12h 20m 9s  | 1/3        | PING OK - Packet loss = 0%, RTA = 2ms |
| <a href="#">AFP</a>       | Ping       | OK        | 02-01-2004 09:55:38 | 0d 13h 40m 29s | 1/3        | PING OK - Packet loss = 0%, RTA = 1ms |
| <a href="#">AGNIPAGE</a>  | Ping       | OK        | 02-01-2004 09:55:27 | 0d 0h 0m 59s   | 1/3        | PING OK - Packet loss = 0%, RTA = 1ms |
| <a href="#">BRTSCHOOL</a> | Ping       | OK        | 02-01-2004 09:54:06 | 1d 18h 7m 39s  | 1/3        | PING OK - Packet loss = 0%, RTA = 8ms |
| <a href="#">Ban-cat</a>   | Ping       | OK        | 02-01-2004 09:56:11 | 0d 22h 44m 39s | 1/3        | PING OK - Packet loss = 0%, RTA = 1ms |

Transferring data from thuldai.mos.com.np...

Current S [root@dhr ? Sun Feb 01, 9:26 PM

[illegible]

## BROWSER



## NOTIFICATIONS

?  
(ticket, ...)



email



SQL

DATA/STATISTICS

**Nagios**  
NETWORK MONITOR

```
define host {
    use host-template
    host_name work-smp
    Alias Back End server
    address 192.168.116.2
    parents work
}
```

CONFIG

SCHEDULE

CHECK  
PING

CHECK  
ORACLE

CHECK  
SPACE

PASSIVE  
CHECKS

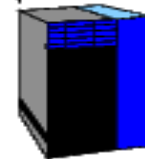


SNMP TRAP  
LISTENER

DISK ERROR



ROUTER



SERVER



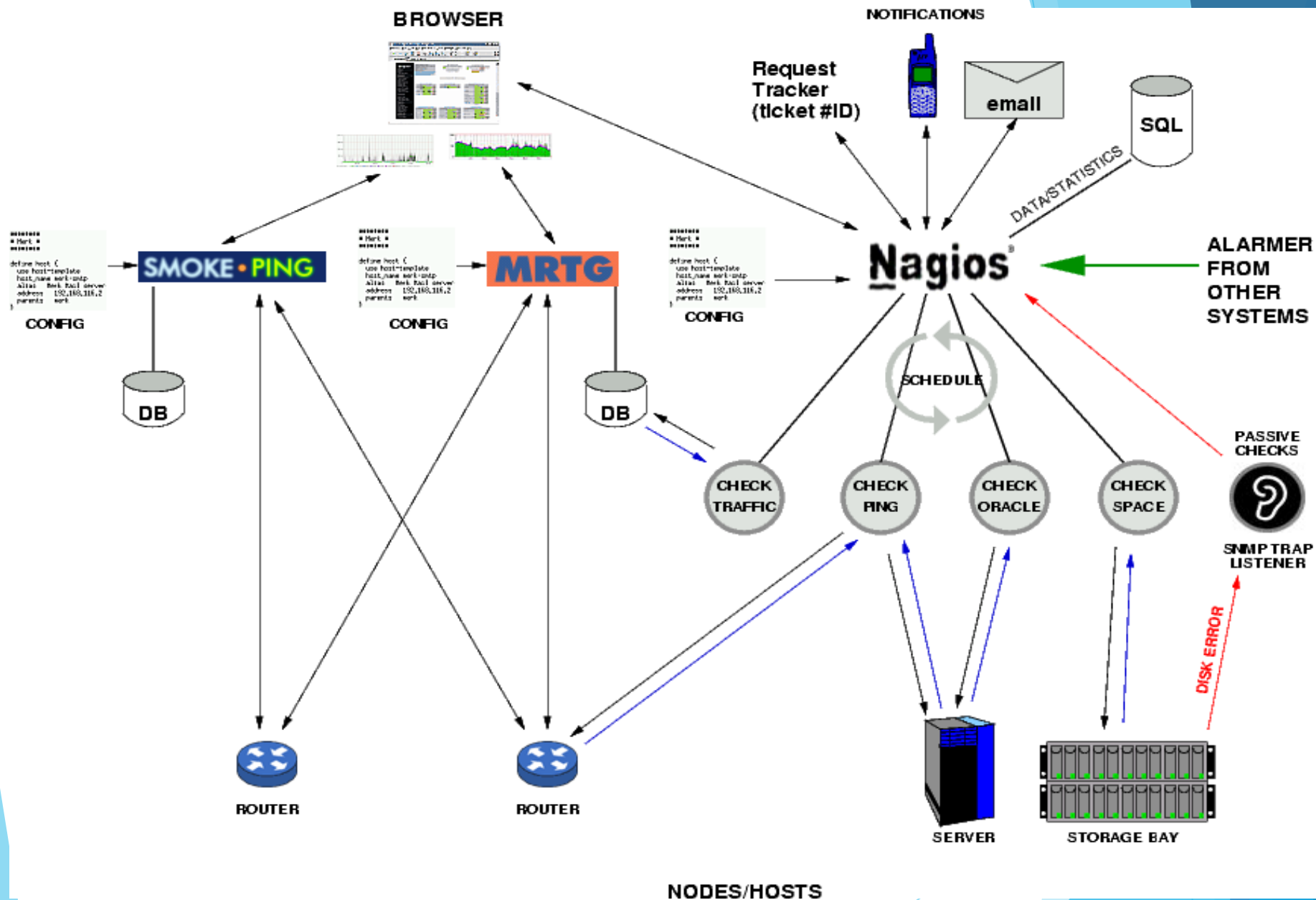
STORAGE BAY

NODES/HOSTS



Hovedstadens Sygehusstøtteselskab

Network Monitoring  
Architecture



# Netflow - *Network Analysis While Preserving Privacy*

- ▶ Privacy concerns today
- ▶ Analyzing traffic usually is done by examining packets – Deep packet inspection
- ▶ Looking at “calling information” can reveal much
- ▶ Can be used as an IDS
- ▶ Can be use as policy enforcement

# NeFlow “Calling Information”

- ▶ Source IP address and port
- ▶ Destination IP address and port
- ▶ Protocol
- ▶ Other data
  - ▶ Timestamps
  - ▶ Number of packets
  - ▶ Bytes
- ▶ Technology used: Netflow

# NetFlow Background

- ▶ Developed by Cisco to:
  - ▶ Characterize traffic
  - ▶ Account for how and where it flows
  - ▶ Help optimize network investment
  - ▶ Traffic engineering/network planning
  - ▶ Provide usage-based billing

# NetFlow Example

Computer A Web browses to Computer B will generate 2 flows:

- ▶ Request flow:

- ▶ A: (TCP) 1.2.3.4:3365 -> 4.3.2.1: 80

- ▶ Reply Flow:

- ▶ B: (TCP) 4.3.2.1:80 -> 1.2.3.4:3365



# NetFlow Typical Record

- ▶ Source and destination IP address
- ▶ Source and destination ports
- ▶ Transport protocol: TCP,UDP, ICMP, etc.
- ▶ Type of service (ToS)
- ▶ Packet and byte counts
- ▶ Start and end timestamps
- ▶ Input and output interface numbers
- ▶ TCP flags
- ▶ Routing information (next-hop address, source autonomous system (AS) number, destination AS number, source prefix mask, destination prefix mask)

# NetFlow Data Cache (1)

- ▶ Available on Cisco routers/switches
- ▶ Available on Juniper routers
- ▶ Cached on devices
- ▶ WARNING! Not all devices are NetFlow-enabled!

# NetFlow Cache Example (2)

| Protocol   | Total      | Flows | Packets | Bytes | Packets | Active (Sec) | Idle (Sec) |
|------------|------------|-------|---------|-------|---------|--------------|------------|
| -----      | Flows      | /Sec  | /Flow   | /Pkt  | /Sec    | /Flow        | /Flow      |
| TCP-Telnet | 3833510    | 0.8   | 10      | 179   | 9.2     | 9.0          | 26.8       |
| TCP-FTP    | 12511306   | 2.9   | 6       | 132   | 19.7    | 6.3          | 16.5       |
| TCP-FTPD   | 1194796    | 0.2   | 544     | 866   | 151.5   | 86.7         | 21.2       |
| TCP-WWW    | 944754736  | 219.9 | 13      | 627   | 2871.0  | 3.2          | 23.7       |
| TCP-SMTP   | 53320030   | 12.4  | 14      | 399   | 185.8   | 6.6          | 19.2       |
| TCP-X      | 913841     | 0.2   | 41      | 631   | 8.9     | 19.2         | 24.5       |
| TCP-BGP    | 1867       | 0.0   | 1       | 49    | 0.0     | 0.5          | 20.5       |
| TCP-NNTP   | 1086658    | 0.2   | 252     | 874   | 63.8    | 15.2         | 26.8       |
| TCP-Frag   | 228697     | 0.0   | 9       | 131   | 0.5     | 6.5          | 25.3       |
| TCP-other  | 2264274585 | 527.1 | 23      | 568   | 12466.6 | 12.9         | 24.4       |
| UDP-DNS    | 231113128  | 53.8  | 2       | 79    | 114.7   | 3.6          | 26.0       |
| UDP-NTP    | 6394017    | 1.4   | 3       | 76    | 5.2     | 9.7          | 27.2       |
| UDP-TFTP   | 13567      | 0.0   | 1       | 95    | 0.0     | 3.1          | 29.3       |
| UDP-Frag   | 211973     | 0.0   | 3165    | 1266  | 156.2   | 116.2        | 27.5       |
| UDP-other  | 1177902953 | 274.2 | 5       | 293   | 1385.8  | 6.1          | 25.8       |
| ICMP       | 103453714  | 24.0  | 2       | 62    | 57.9    | 3.8          | 26.0       |
| IGMP       | 726        | 0.0   | 2       | 300   | 0.0     | 3.5          | 29.1       |
| IPINIP     | 1          | 0.0   | 1       | 40    | 0.0     | 0.0          | 27.2       |

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# NetFlow Privacy

- ▶ While the contents of the packet are not recorded, the calling information can still be a concern.
- ▶ However, with virtual servers, it is impossible to know the true destination
- ▶ Mostly it can only be used as verification that something occurred.

# Network Management Platforms

# ITIL in Hewlett Packard (HP)

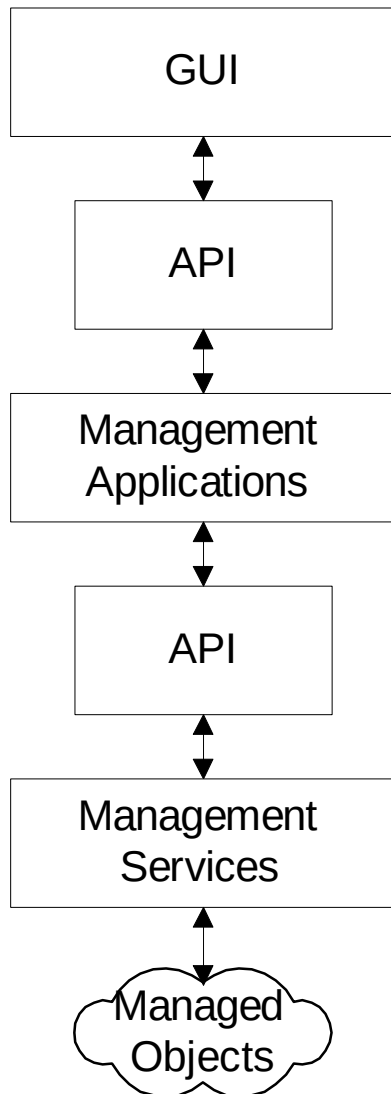
- ▶ An active supporter of ITIL since 1995
- ▶ The first major corporate sponsor of the IT Service Management Forum in the United States.
- ▶ The first major vendor to adopt and the highest number of ITIL-certified consultants across the globe of the ITIL standard since 1996.
- ▶ HP built ITIL standards into **HP OpenView software** and **ITSM** enables organizations to improve IT at different levels- from problem management to change management to service level management. ([www.itsmf.com](http://www.itsmf.com))

# HP OpenView Platform (1)

- ▶ HP OpenView is not one product but a range of over 50 software products dedicated to administrative problem-solving in IT departments.
- ▶ It isn't a single piece of software but a whole suite of different tools to handle the administration of system-performance and availability.
- ▶ The suite has been assembled piecemeal over the decades by HP buying up different pieces of software and consolidating them under the OpenView (now rebranded as HP-OM) umbrella.



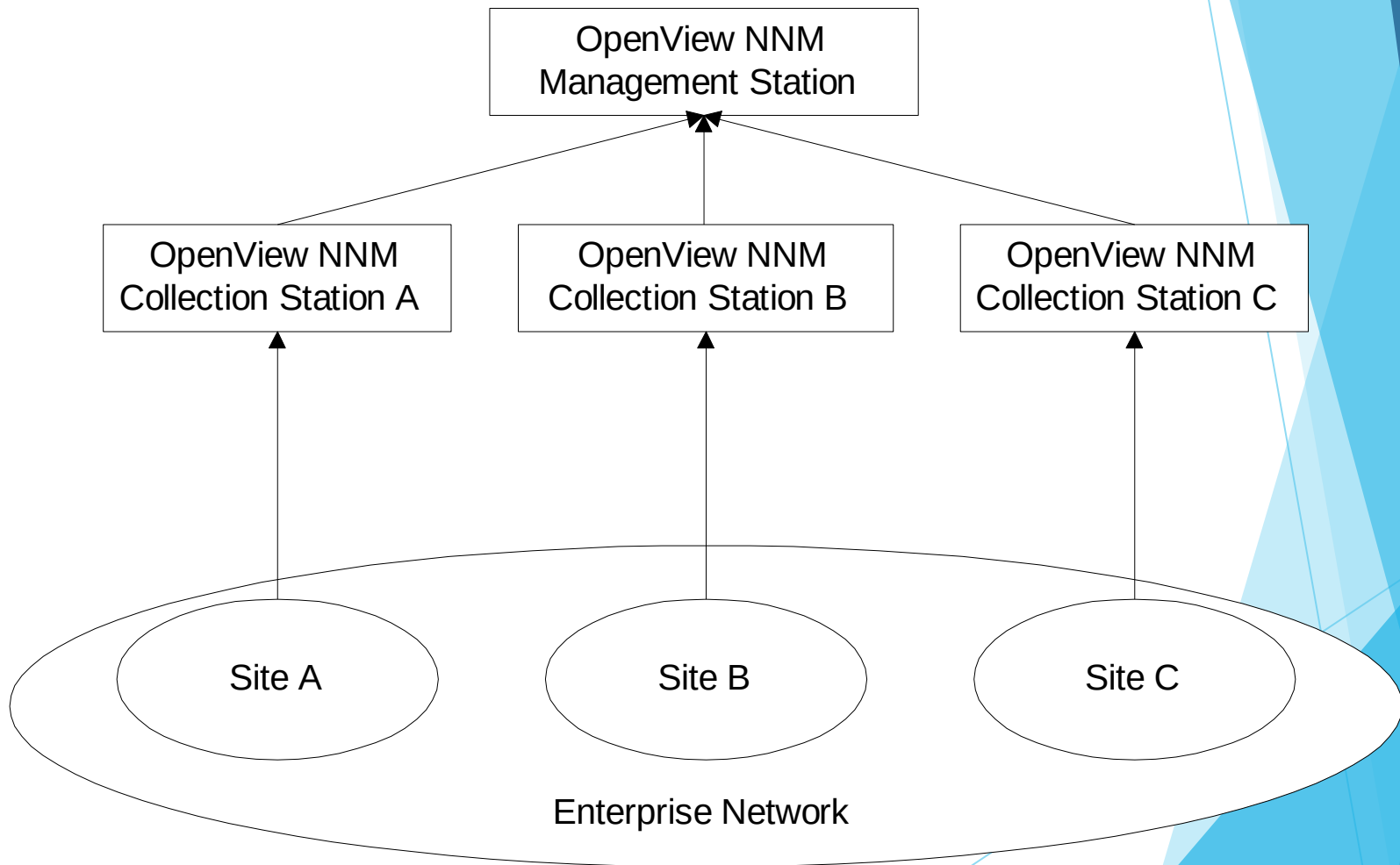
# HP OpenView Platform (2)



- Open, modular, and distributed architecture
- Object oriented design
- Open API-based architecture
- Easy vendor-specific NMS integration by 3rd party



# Distributed OpenView Network Node Managers (NNMs)



# Further Management Platforms

- ▶ Tivoli TME Management Platform
- ▶ Cabletron Spectrum Management Platform
- ▶ Patrol Management System

Thank you for your attention!